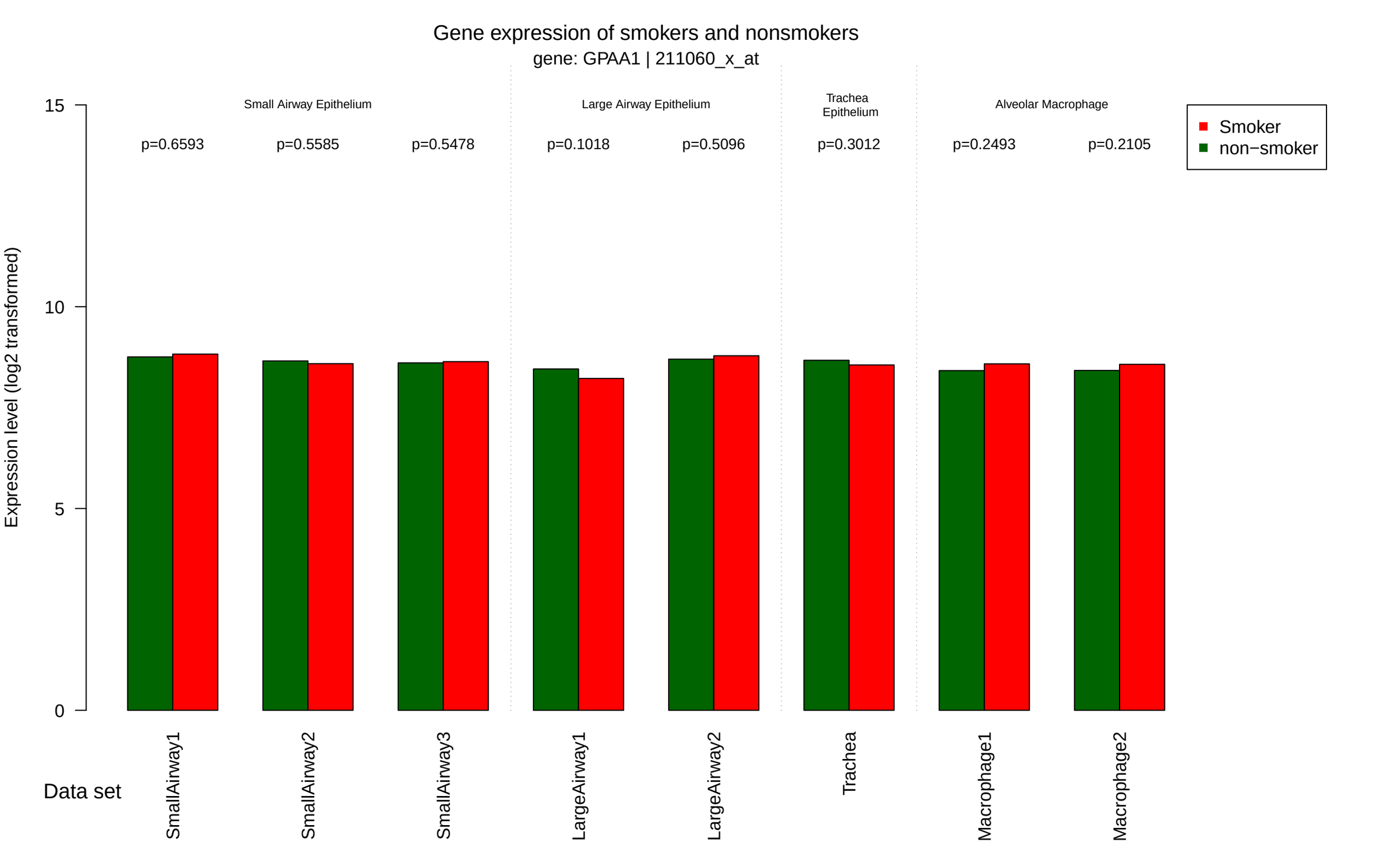
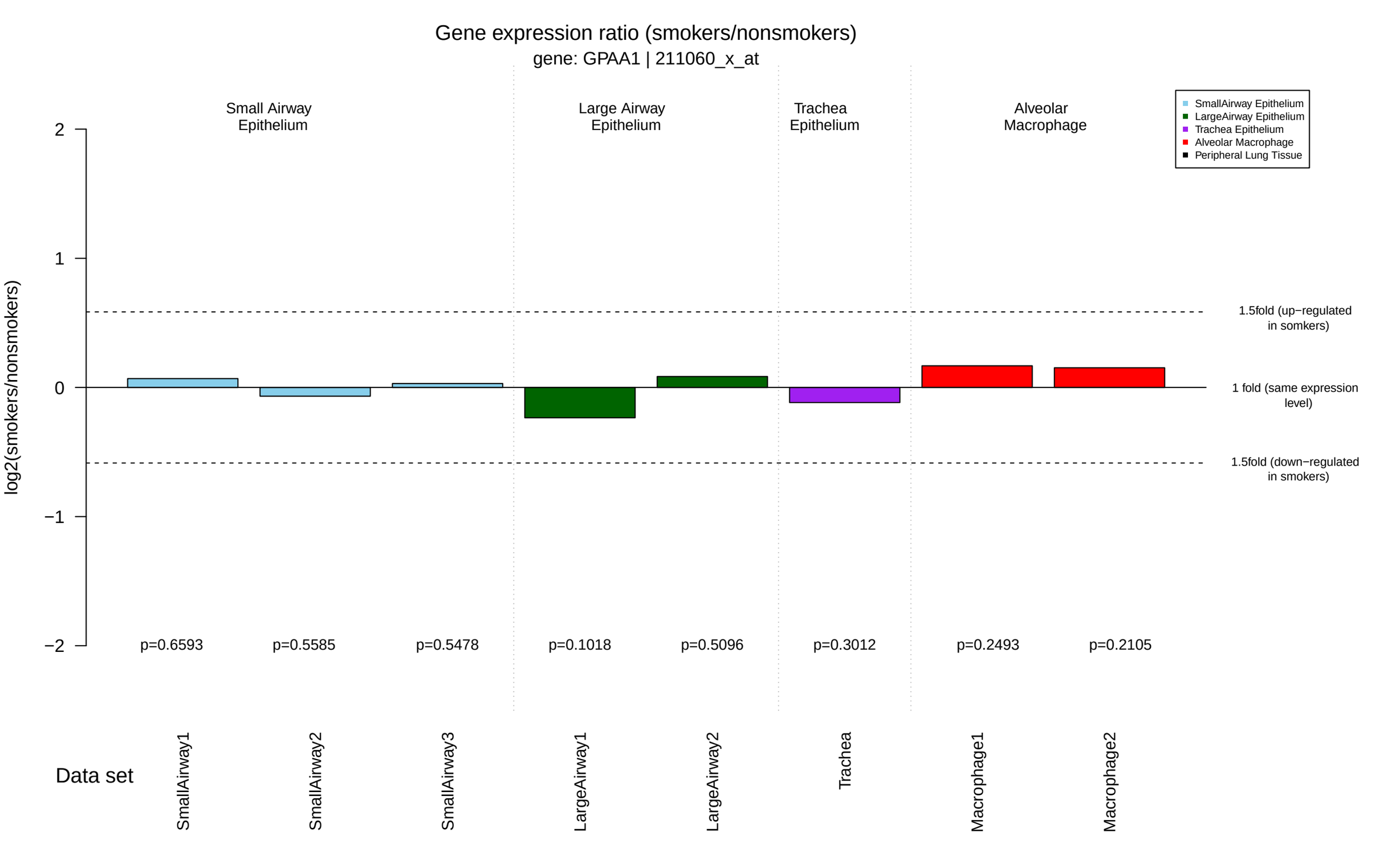
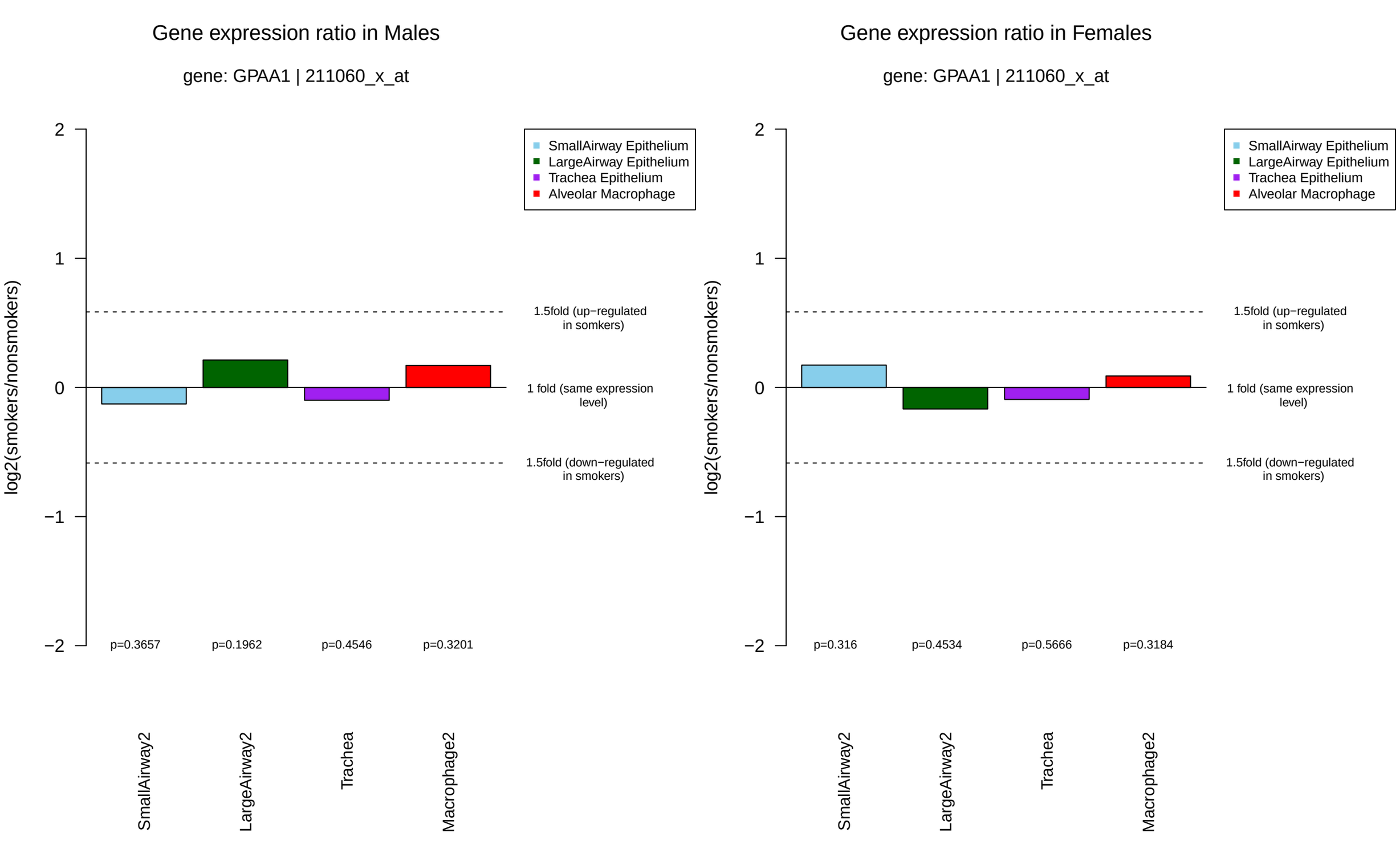
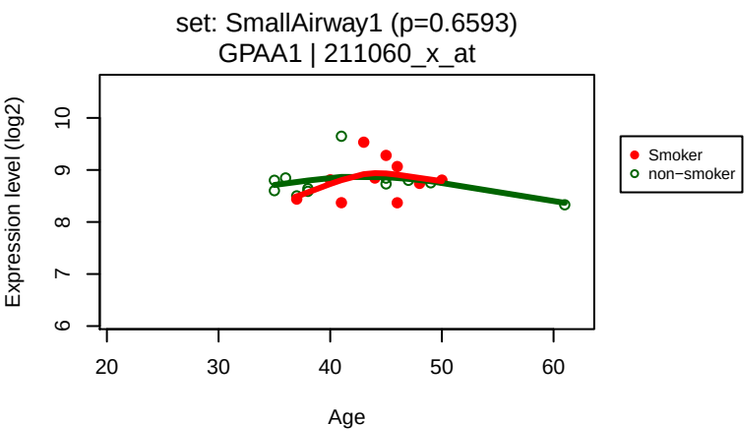
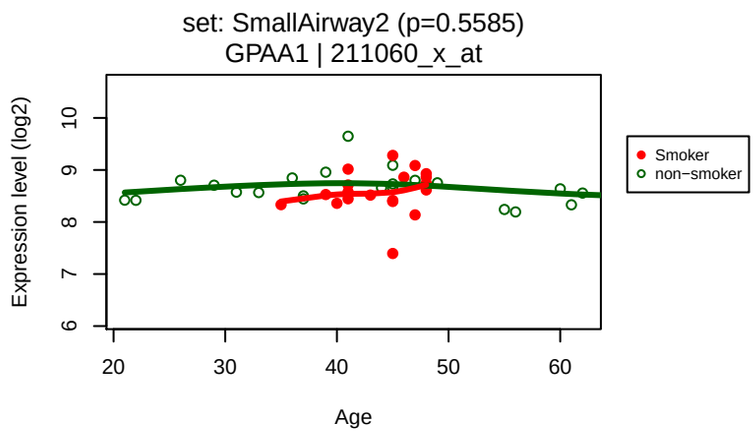
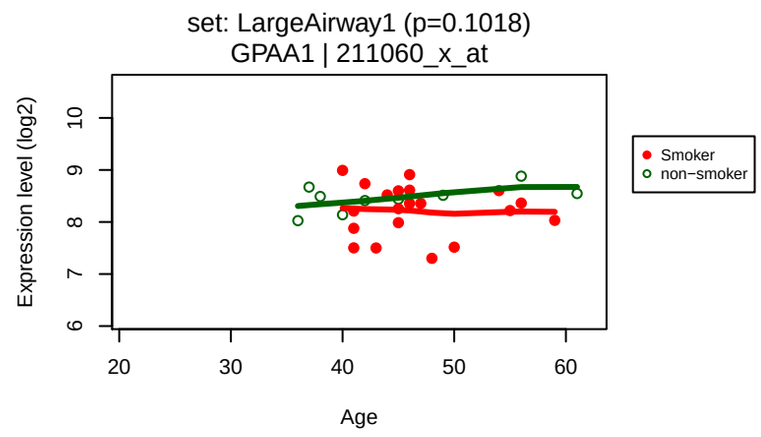

Gene expression of smokers and nonsmokers
gene: GPAA1 | 211060_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6593
p=0.5585
p=0.5478
p=0.1018
p=0.5096
p=0.3012
p=0.2493
p=0.2105
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

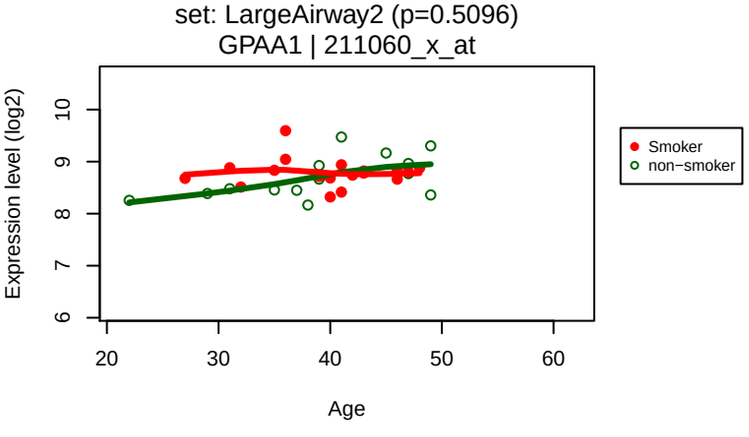

Gene expression ratio (smokers/nonsmokers)
gene: GPAA1 | 211060_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6593
p=0.5585
p=0.5478
p=0.1018
p=0.5096
p=0.3012
p=0.2493
p=0.2105
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

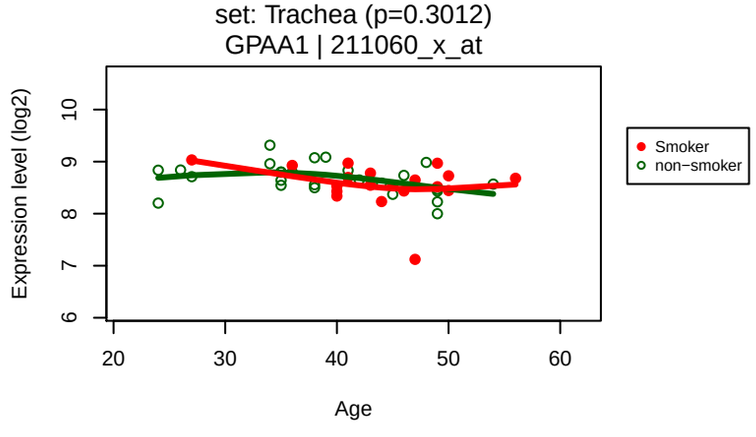

Gene expression ratio in Males
gene: GPAA1 | 211060_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3657
p=0.1962
p=0.4546
p=0.3201
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPAA1 | 211060_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.316
p=0.4534
p=0.5666
p=0.3184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

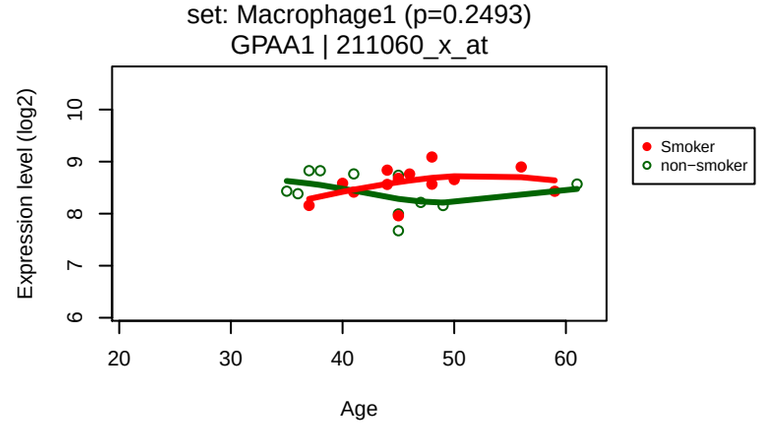

set: SmallAirway1 (p=0.6593)
GPAA1 | 211060_x_at
Expression level (log2)
Age
Smoker
non−smoker

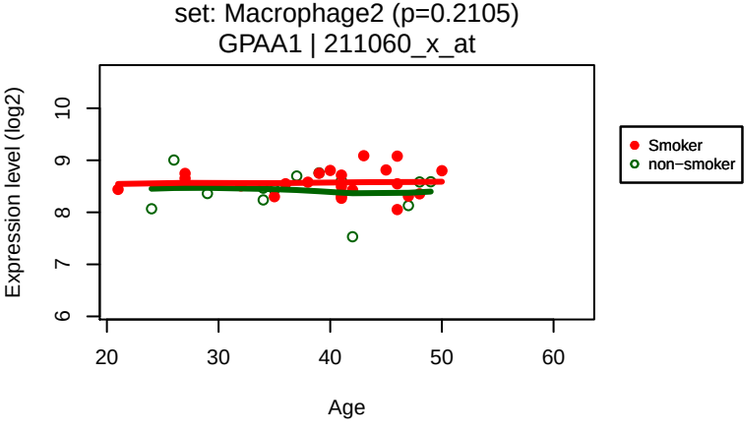

set: SmallAirway2 (p=0.5585)
GPAA1 | 211060_x_at
Expression level (log2)
Age
Smoker
non−smoker

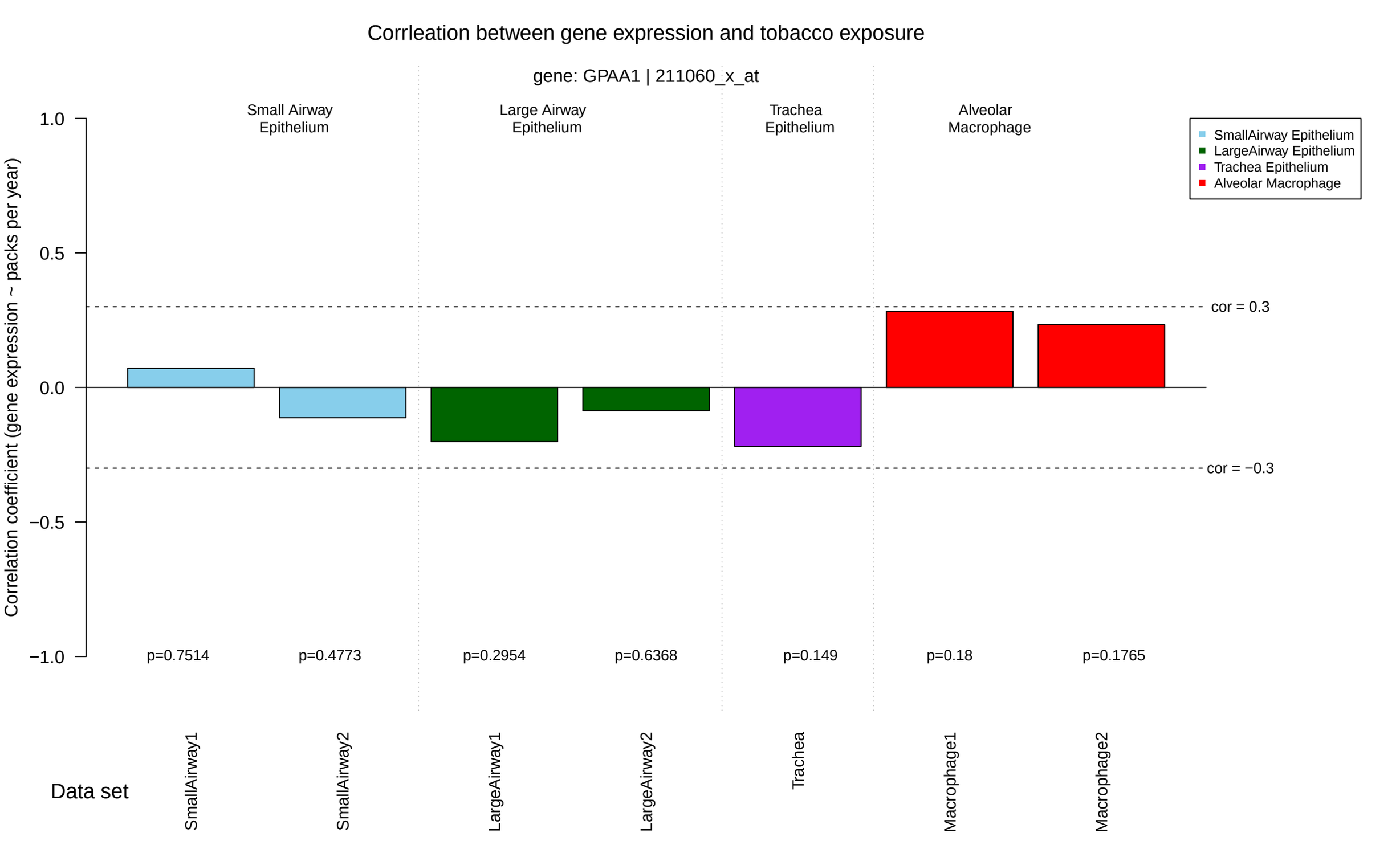

set: LargeAirway1 (p=0.1018)
GPAA1 | 211060_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5096)
GPAA1 | 211060_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3012)
GPAA1 | 211060_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2493)
GPAA1 | 211060_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2105)
GPAA1 | 211060_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPAA1 | 211060_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7514
p=0.4773
p=0.2954
p=0.6368
p=0.149
p=0.18
p=0.1765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2